Formal Verification: Quasar Protocol: BLS + Ringtail Hybrid Finality

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Abstract

We formalize the Quasar protocol, the quantum finality engine that produces dual-signed (BLS + Ringtail) aggregate blocks. We prove BLS aggregation soundness, quorum intersection, dual signature requirements, and epoch monotonicity.

1 Introduction

Quasar operates at the protocol level, collecting blocks from all Lux chains (P, C, X, A, etc.) and producing quantum-final aggregate blocks. Each block carries both BLS (fast, classical) and Ringtail (post-quantum, lattice-based) signatures, ensuring quantum resistance.

2 Definitions

Definition 1 (BLSSig / RingtailSig)**.** *Abstract signature types with validity flags.*

Definition 2 (QuasarSig)**.** *Dual signature: BLS + Ringtail components.*

Definition 3 (QuantumBlock)**.** *A quantum block with height, source block hashes, and dual signature.*

Definition 4 (QuasarState)**.** *State: quantum height and list of finalized quantum blocks.*

3 Theorems and Axioms

Axiom 5 (`bls_aggregation_sound`)**.** *Aggregation of valid BLS signatures produces a valid aggregate (bilinear pairing property).*

Theorem 6 (`height_monotone`)**.** *Quantum height is strictly increasing when a new block is added.*

Proof. By `simp` on `addBlock`. □

Theorem 7 (`bls_quorum_intersection`)**.** *Two quorums of $\geq 2f+1$ from n validators with $3f < n$ overlap: $q_1 + q_2 > n$.*

Proof. By `omega`. □

Theorem 8 (`quantum_finality_requires_both`)**.** *A valid quantum signature requires both BLS and Ringtail components to be valid.*

Proof. Decompose the Boolean conjunction. □

4 Lean Source

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  Quasar Protocol Formal Model

  Models protocol/quasar/ -- BLS+Ringtail hybrid quantum consensus.

  Quasar is the finality engine. It collects blocks from all chains
  and produces quantum-final aggregate blocks with dual signatures:
  - BLS (classical, fast): 96-byte aggregated signatures
  - Ringtail (post-quantum, larger): ~4KB lattice-based threshold signatures

  Maps to:
  - quasar/bls.go: BLS signature aggregation (quorum = 2f+1)
  - quasar/core.go: Quasar struct, chain buffers, quantum finality
  - quasar/engine.go: Engine implementation
  - quasar/horizon.go: Event horizon (pending blocks)

  Properties:
  - BLS aggregation correctness: agg_verify(agg_sig, agg_pk, msg) iff all
      individual sigs valid
  - Quorum threshold: 2f+1 signatures required for finality
  - Dual signature: quantum block has both BLS and Ringtail sigs
  - Epoch monotonicity: quantum height only increases
-/

import Mathlib.Data.Nat.Defs
import Mathlib.Tactic

namespace Protocol.Quasar

/-- Signature types -/
structure BLSSig where
  bytes : List UInt8
  valid : Bool

structure RingtailSig where
  bytes : List UInt8
  valid : Bool

/-- Quasar signature = BLS + Ringtail (models quasar.QuasarSig) -/
structure QuasarSig where
  bls      : BLSSig
  ringtail : RingtailSig

/-- Quantum block (models quasar.QuantumBlock) -/
structure QuantumBlock where
  height       : Nat
  sourceBlocks : List Nat   -- block hashes
  signature    : QuasarSig

/-- Quasar state -/
structure QuasarState where
  quantumHeight : Nat
```

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  finalized     : List QuantumBlock

/-- BLS aggregation: aggregate of valid sigs is valid -/
axiom bls_aggregation_sound :
   (sigs : List BLSSig),
    ( s  sigs, s.valid = true) →
    sigs  [] →
    -- Aggregated signature is valid (bilinear pairing property)
     (agg : BLSSig), agg.valid = true

/-- Quantum height is monotonically increasing -/
def addBlock (s : QuasarState) (b : QuantumBlock)
    (h : b.height > s.quantumHeight) : QuasarState :=
  { quantumHeight := b.height
  , finalized := b :: s.finalized }

theorem height_monotone (s : QuasarState) (b : QuantumBlock)
    (h : b.height > s.quantumHeight) :
    (addBlock s b h).quantumHeight > s.quantumHeight := by
  simp [addBlock]
  exact h

/-- Quorum intersection for BLS signatures:
    Two quorums of 2f+1 from n=3f+1 validators share f+1 members.
    At least one shared member is honest. -/
theorem bls_quorum_intersection
    (n f : Nat) (hf : 3 * f < n)
    (q1 q2 : Nat)
    (hq1 : q1  2 * f + 1) (hq2 : q2  2 * f + 1) :
    q1 + q2 > n := by
  omega

/-- Dual signature validity: both components must be valid -/
def isValidQuantumSig (qs : QuasarSig) : Bool :=
  qs.bls.valid && qs.ringtail.valid

/-- A finalized block has both BLS and Ringtail valid -/
theorem quantum_finality_requires_both
    (qs : QuasarSig) (h : isValidQuantumSig qs = true) :
    qs.bls.valid = true  qs.ringtail.valid = true := by
  simp [isValidQuantumSig] at h
  exact h

end Protocol.Quasar
```

5 References

Source code: `https://github.com/luxfi/proofs/blob/main/lean/Protocol/Quasar.lean`
White papers: `https://papers.lux.network`
Related white papers:

- `lux-quasar-consensus.tex`
- `lux-quantum-consensus.tex`